Time Series HW2

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1 Task 6.1

We consider a Moving Average process of order 2 (MA(2)) defined as:

$$X_t = \varepsilon_t + 0.5\varepsilon_{t-1} - 0.3\varepsilon_{t-2}, \quad \varepsilon_t \sim \mathcal{N}(0,1) \tag{1}$$

1.1 Theoretical Autocovariance Function $\gamma(k)$

For a general MA(2) process, the autocovariance function is given by:

$$\gamma(k) = \mathbb{E}[X_t X_{t-k}] = \begin{cases} \sigma^2(1+\theta_1^2+\theta_2^2) & \text{for } k=0 \\ \sigma^2(\theta_1+\theta_1\theta_2) & \text{for } k=1 \\ \sigma^2\theta_2 & \text{for } k=2 \\ 0 & \text{for } k \geq 3 \end{cases} \tag{2}$$

Substituting our parameter values $\theta_1 = 0.5$, $\theta_2 = -0.3$, $\sigma = 1$:

1.2 Calculations

$$\begin{aligned} \gamma(0) &= 1 \times (1 + 0.5^2 + (-0.3)^2) \\ &= 1 + 0.25 + 0.09 \\ &= 1.34 \end{aligned} \tag{3}$$

$$\begin{aligned} \gamma(1) &= 1 \times (0.5 + 0.5 \times (-0.3)) \\ &= 0.5 - 0.15 \\ &= 0.35 \end{aligned} \tag{4}$$

$$\begin{aligned} \gamma(2) &= 1 \times (-0.3) \\ &= -0.3 \end{aligned} \tag{5}$$

$$\gamma(k) = 0 \quad \text{for all } k \geq 3$$

1.3 Theoretical Autocorrelation Function $\rho(k)$

The autocorrelation function is defined as:

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} \tag{6}$$

1.4 Calculations

$$\rho(0) = \frac{\gamma(0)}{\gamma(0)} = 1.0000 \tag{7}$$

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{0.35}{1.34} \approx 0.2612 \tag{8}$$

$$\rho(2) = \frac{\gamma(2)}{\gamma(0)} = \frac{-0.30}{1.34} \approx -0.2239 \tag{9}$$

$$\rho(k) = 0 \quad \text{for all } k \geq 3$$

1.5 Summary Table

Table 1: Theoretical Autocorrelation Function

k	Formula	Value
0	1	1.0000
1	$\frac{\theta_1 + \theta_1\theta_2}{1 + \theta_1^2 + \theta_2^2}$	0.2612
2	$\frac{\theta_2}{1 + \theta_1^2 + \theta_2^2}$	-0.2239
3	0	0.0000
4	0	0.0000
5	0	0.0000
$\vdots$	$\vdots$	$\vdots$

1.6 Mean and Variance

$$\mathbb{E}[X_t] = 0 \quad (\text{since } \mathbb{E}[\varepsilon_t] = 0) \tag{10}$$

$$\text{Var}(X_t) = \gamma(0) = 1.34 \tag{11}$$

1.7 Simulation Results

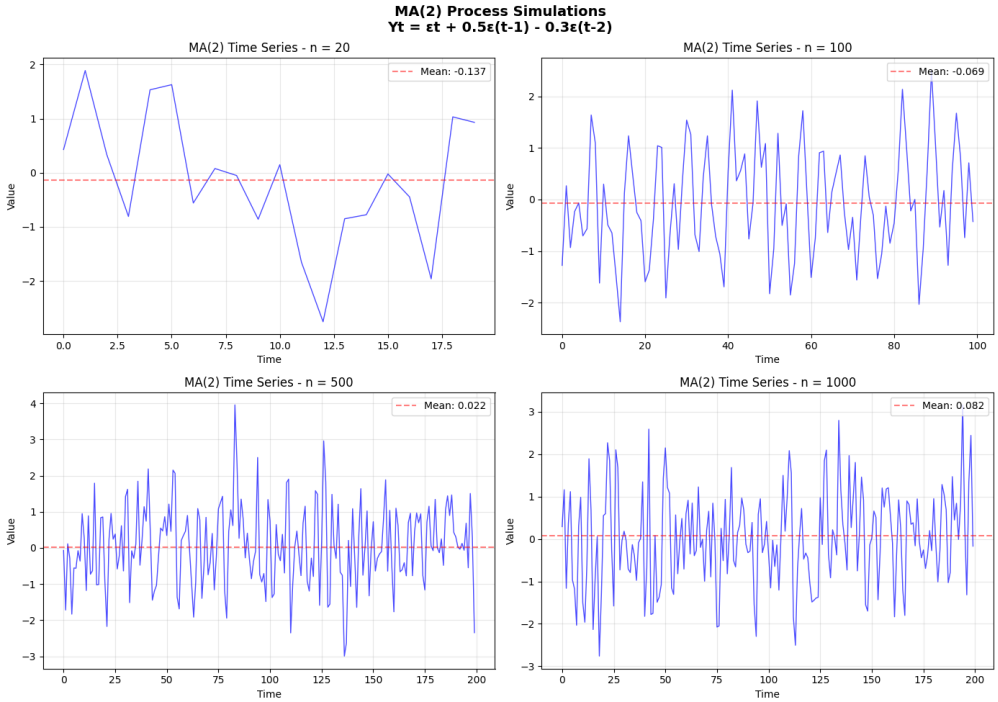

Figure 1: MA(2) process

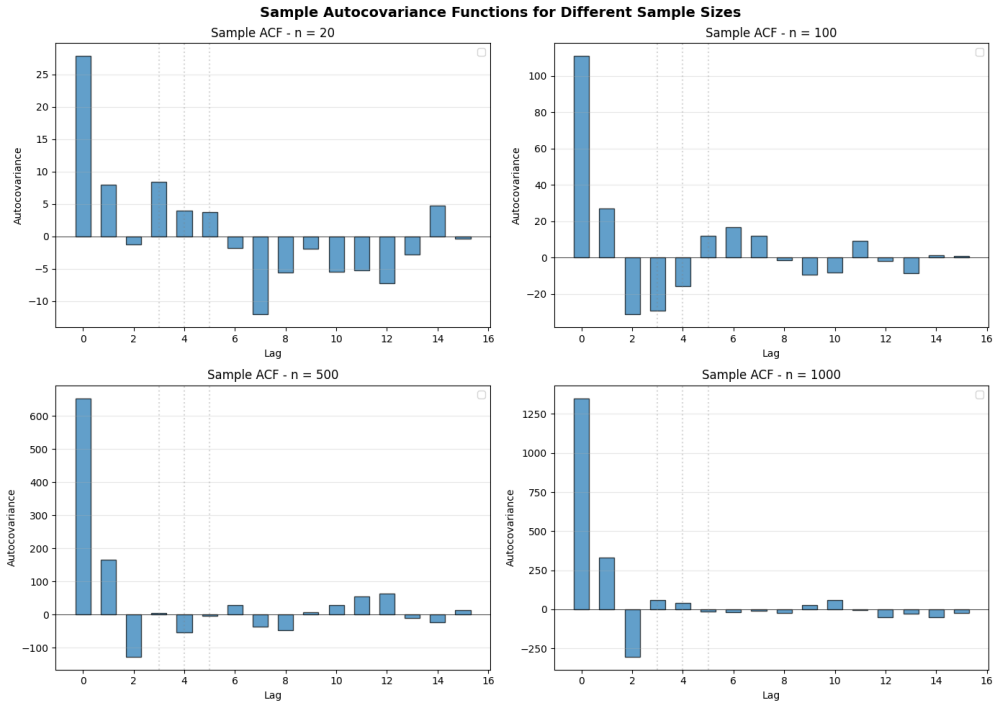

Figure 2: Sample autocovariance function (ACF) of the simulated MA(2) process

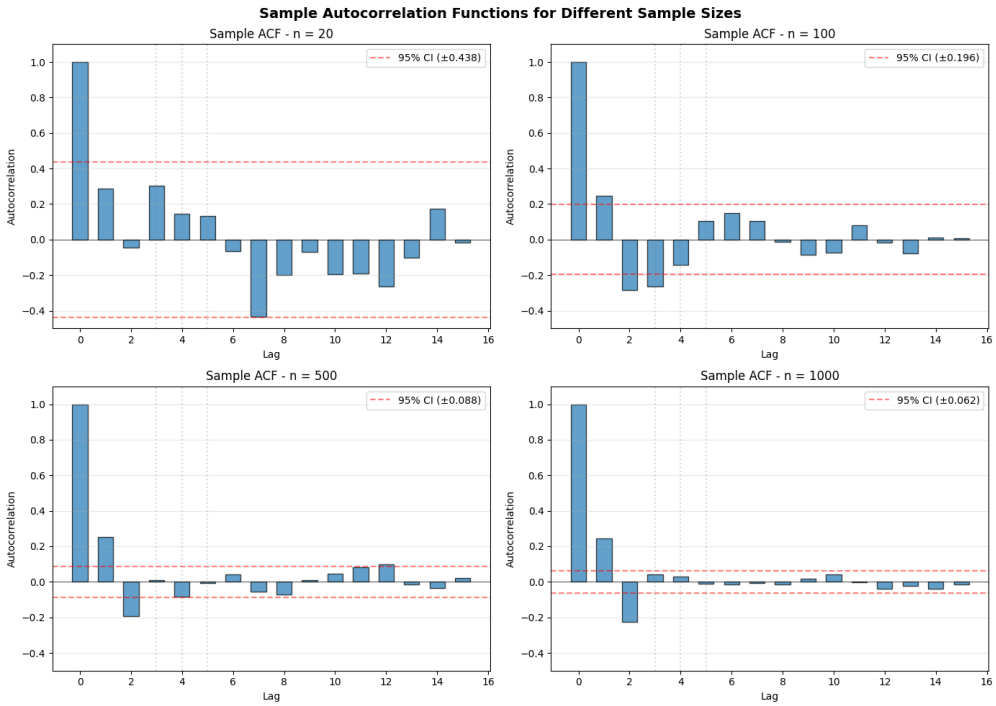

Figure 3: Sample autocorrelation function of the simulated MA(2) process.

Significance level: α = 0.05

Sample Size	Lags	Q-statistic	p-value	Decision
20	5	5.4035	0.368649	Fail to reject H_0
20	10	15.2646	0.122708	Fail to reject H_0
20	15	23.6839	0.070660	Fail to reject H_0
100	5	25.2086	0.000127	REJECT H_0
100	10	30.3666	0.000746	REJECT H_0
100	15	31.8856	0.006673	REJECT H_0
500	5	54.8786	0.000000	REJECT H_0
500	10	61.0080	0.000000	REJECT H_0
500	15	70.3836	0.000000	REJECT H_0
1000	5	113.9085	0.000000	REJECT H_0
1000	10	116.6815	0.000000	REJECT H_0
1000	15	120.4218	0.000000	REJECT H_0

Figure 4: Autocorrelation tests

2 Task 6.2

We consider an Autoregressive process of order 2 (AR(2)) defined as:

$$X_t = 0.5X_{t-1} + 0.3X_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1) \tag{12}$$

2.1 Stationarity Conditions

For an AR(2) process to be stationary, the roots of the characteristic equation must lie outside the unit circle. The characteristic equation is:

$$1 - \phi_1 z - \phi_2 z^2 = 0 \tag{13}$$

With our parameters $\phi_1 = 0.5$, $\phi_2 = 0.3$:

$$\begin{aligned} 1 - 0.5z - 0.3z^2 &= 0 \\ -0.3z^2 - 0.5z + 1 &= 0 \\ 0.3z^2 + 0.5z - 1 &= 0 \end{aligned} \tag{14}$$

The roots are:

$$\begin{aligned} z_{1,2} &= \frac{-0.5 \pm \sqrt{0.5^2 + 4 \times 0.3 \times 1}}{2 \times 0.3} \\ &= \frac{-0.5 \pm \sqrt{0.25 + 1.2}}{0.6} \\ &= \frac{-0.5 \pm \sqrt{1.45}}{0.6} \\ &= \frac{-0.5 \pm 1.2042}{0.6} \end{aligned}$$

$$z_1 = \frac{-0.5 + 1.2042}{0.6} = \frac{0.7042}{0.6} \approx 1.1737 \tag{15}$$

$$z_2 = \frac{-0.5 - 1.2042}{0.6} = \frac{-1.7042}{0.6} \approx -2.8403 \tag{16}$$

Since both roots $|z_1| > 1$ and $|z_2| > 1$, the AR(2) process is stationary.

2.2 Theoretical Autocovariance Function $\gamma(k)$

For an AR(2) process, the autocovariance function satisfies the Yule-Walker equations:

$$\gamma(0) = \phi_1\gamma(1) + \phi_2\gamma(2) + \sigma^2 \tag{17}$$

$$\gamma(1) = \phi_1\gamma(0) + \phi_2\gamma(1) \tag{18}$$

$$\gamma(2) = \phi_1\gamma(1) + \phi_2\gamma(0) \tag{19}$$

For $k \geq 3$, we have the recurrence relation:

$$\gamma(k) = \phi_1\gamma(k-1) + \phi_2\gamma(k-2) \quad \text{for } k \geq 3 \tag{20}$$

2.3 Solving the Yule-Walker Equations

From equation (18):

$$\begin{aligned} \gamma(1) &= \phi_1\gamma(0) + \phi_2\gamma(1) \\ \gamma(1) - \phi_2\gamma(1) &= \phi_1\gamma(0) \\ (1 - \phi_2)\gamma(1) &= \phi_1\gamma(0) \\ \gamma(1) &= \frac{\phi_1}{1 - \phi_2}\gamma(0) \end{aligned} \tag{21}$$

From equation (19):

$$\gamma(2) = \phi_1\gamma(1) + \phi_2\gamma(0) \tag{22}$$

Substituting (21) into (17):

$$\begin{aligned}\gamma(0) &= \phi_1\left(\frac{\phi_1}{1-\phi_2}\gamma(0)\right) + \phi_2\gamma(2) + \sigma^2 \\ &= \frac{\phi_1^2}{1-\phi_2}\gamma(0) + \phi_2\gamma(2) + \sigma^2 \end{aligned} \tag{23}$$

Also, from (22) and (21):

$$\begin{aligned}\gamma(2) &= \phi_1\left(\frac{\phi_1}{1-\phi_2}\gamma(0)\right) + \phi_2\gamma(0) \\ &= \left(\frac{\phi_1^2}{1-\phi_2} + \phi_2\right)\gamma(0) \end{aligned} \tag{24}$$

Substituting (24) into (23):

$$\gamma(0) = \frac{\phi_1^2}{1-\phi_2}\gamma(0) + \phi_2\left(\frac{\phi_1^2}{1-\phi_2} + \phi_2\right)\gamma(0) + \sigma^2$$

$$\gamma(0)\left[1 - \frac{\phi_1^2}{1-\phi_2} - \phi_2\left(\frac{\phi_1^2}{1-\phi_2} + \phi_2\right)\right] = \sigma^2$$

Simplifying and solving for $\gamma(0)$:

$$\gamma(0) = \frac{(1-\phi_2)\sigma^2}{(1+\phi_2)[(1-\phi_2)^2 - \phi_1^2]} \tag{25}$$

2.4 Calculations with $\phi_1 = 0.5$, $\phi_2 = 0.3$, $\sigma = 1$

$$\begin{aligned}\gamma(0) &= \frac{(1-0.3)\times 1^2}{(1+0.3)[(1-0.3)^2 - 0.5^2]} \\ &= \frac{0.7}{1.3\times(0.7^2 - 0.25)} \\ &= \frac{0.7}{1.3\times(0.49 - 0.25)} \\ &= \frac{0.7}{1.3\times 0.24} \\ &= \frac{0.7}{0.312} \\ &= 2.2436 \end{aligned} \tag{26}$$

$$\begin{aligned}\gamma(1) &= \frac{0.5}{1-0.3}\times 2.2436 \\ &= \frac{0.5}{0.7}\times 2.2436 \\ &= 0.7143\times 2.2436 \\ &= 1.6026 \end{aligned} \tag{27}$$

$$\begin{aligned}\gamma(2) &= 0.5\times 1.6026 + 0.3\times 2.2436 \\ &= 0.8013 + 0.6731 \\ &= 1.4744 \end{aligned} \tag{28}$$

For $k = 3$:

$$\begin{aligned}\gamma(3) &= 0.5 \times \gamma(2) + 0.3 \times \gamma(1) \\ &= 0.5 \times 1.4744 + 0.3 \times 1.6026 \\ &= 0.7372 + 0.4808 \\ &= 1.2180 \end{aligned} \tag{29}$$

For $k = 4$:

$$\begin{aligned}\gamma(4) &= 0.5 \times \gamma(3) + 0.3 \times \gamma(2) \\ &= 0.5 \times 1.2180 + 0.3 \times 1.4744 \\ &= 0.6090 + 0.4423 \\ &= 1.0513 \end{aligned} \tag{30}$$

For $k = 5$:

$$\begin{aligned}\gamma(5) &= 0.5 \times \gamma(4) + 0.3 \times \gamma(3) \\ &= 0.5 \times 1.0513 + 0.3 \times 1.2180 \\ &= 0.5256 + 0.3654 \\ &= 0.8910 \end{aligned} \tag{31}$$

2.5 Summary Table for Autocovariance Function

Table 2: Theoretical Autocovariance Function for AR(2)

k	Formula	Value
0	$\dfrac{(1-\phi_2)\sigma^2}{(1+\phi_2)[(1-\phi_2)^2-\phi_1^2]}$	2.2436
1	$\dfrac{\phi_1}{1-\phi_2}\gamma(0)$	1.6026
2	$\phi_1\gamma(1)+\phi_2\gamma(0)$	1.4744
3	$\phi_1\gamma(2)+\phi_2\gamma(1)$	1.2180
4	$\phi_1\gamma(3)+\phi_2\gamma(2)$	1.0513
5	$\phi_1\gamma(4)+\phi_2\gamma(3)$	0.8910
$\vdots$	$\vdots$	$\vdots$

2.6 Theoretical Autocorrelation Function $\rho(k)$

The autocorrelation function is defined as:

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} \tag{32}$$

2.7 Calculations

$$\rho(0) = \frac{\gamma(0)}{\gamma(0)} = 1.0000 \tag{33}$$

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{1.6026}{2.2436} \approx 0.7143 \tag{34}$$

$$\rho(2) = \frac{\gamma(2)}{\gamma(0)} = \frac{1.4744}{2.2436} \approx 0.6574 \tag{35}$$

$$\rho(3) = \frac{\gamma(3)}{\gamma(0)} = \frac{1.2180}{2.2436} \approx 0.5430 \tag{36}$$

$$\rho(4) = \frac{\gamma(4)}{\gamma(0)} = \frac{1.0513}{2.2436} \approx 0.4687 \tag{37}$$

$$\rho(5) = \frac{\gamma(5)}{\gamma(0)} = \frac{0.8910}{2.2436} \approx 0.3973 \tag{38}$$

2.8 Summary Table for Autocorrelation Function

Table 3: Theoretical Autocorrelation Function for AR(2)

k	Value	Recurrence Relation
0	1.0000	$\rho(0) = 1$
1	0.7143	$\rho(1) = \frac{\phi_1}{1-\phi_2}$
2	0.6574	$\rho(2) = \phi_1\rho(1) + \phi_2$
3	0.5430	$\rho(3) = \phi_1\rho(2) + \phi_2\rho(1)$
4	0.4687	$\rho(4) = \phi_1\rho(3) + \phi_2\rho(2)$
5	0.3973	$\rho(5) = \phi_1\rho(4) + \phi_2\rho(3)$
$\vdots$	$\vdots$	$\rho(k) = \phi_1\rho(k-1) + \phi_2\rho(k-2)$

2.9 Mean and Variance

$$\mathbb{E}[X_t] = 0 \quad \text{(since } \mathbb{E}[\varepsilon_t] = 0 \text{ and no constant term)} \tag{39}$$

$$\mathrm{Var}(X_t) = \gamma(0) = 2.2436 \tag{40}$$

2.10 Simulation Results

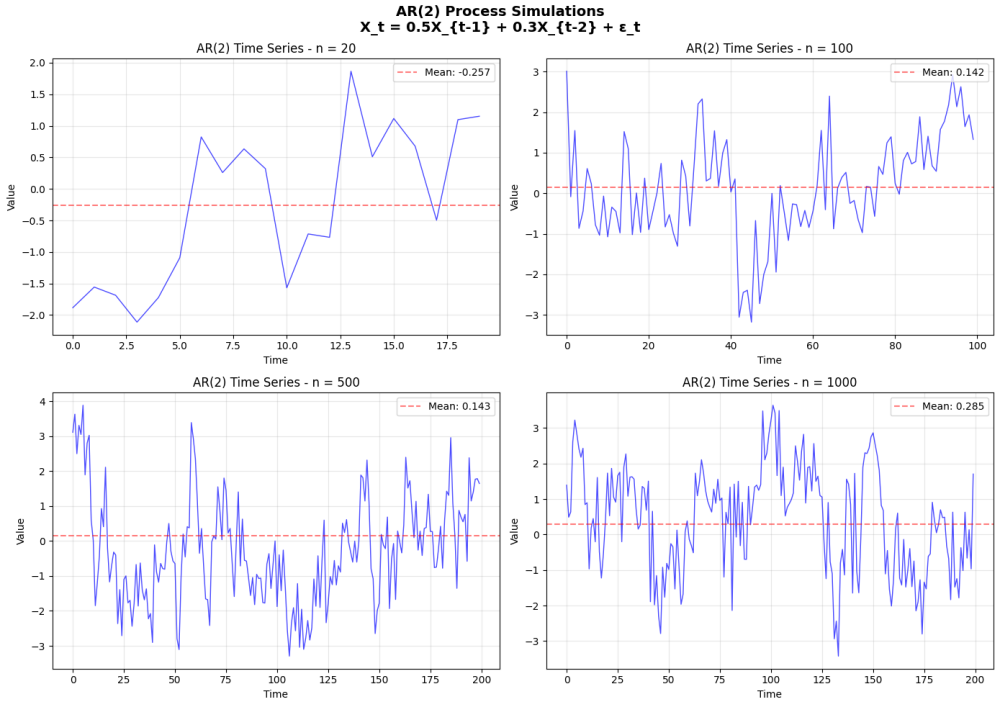

Figure 5: AR(2) process

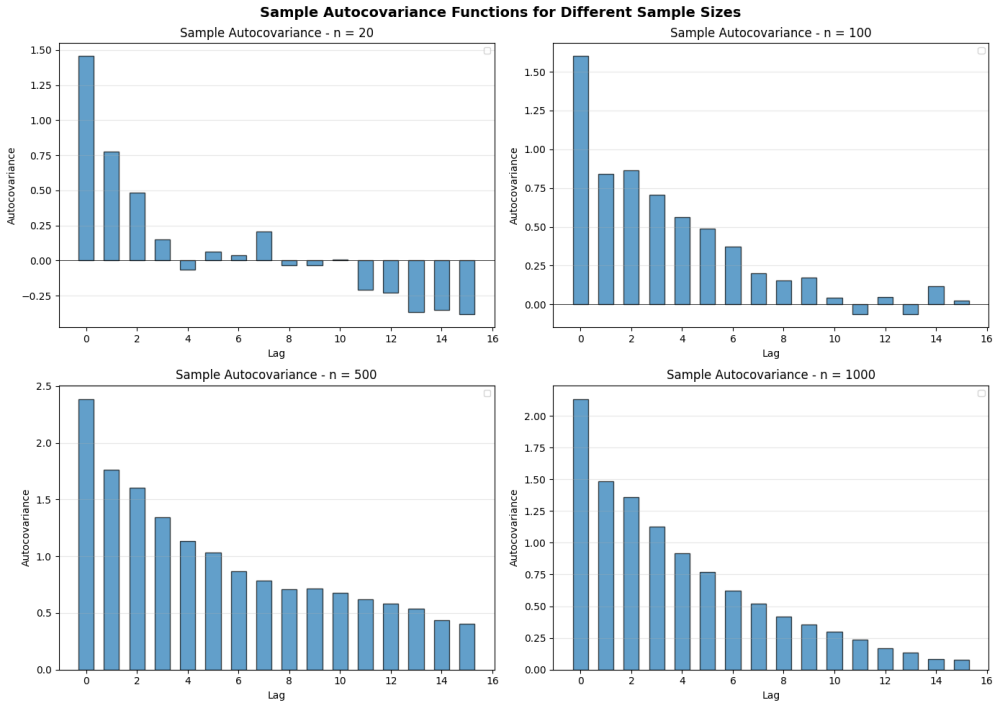

Figure 6: Sample autocovariance function (ACF) of the simulated AR(2) process

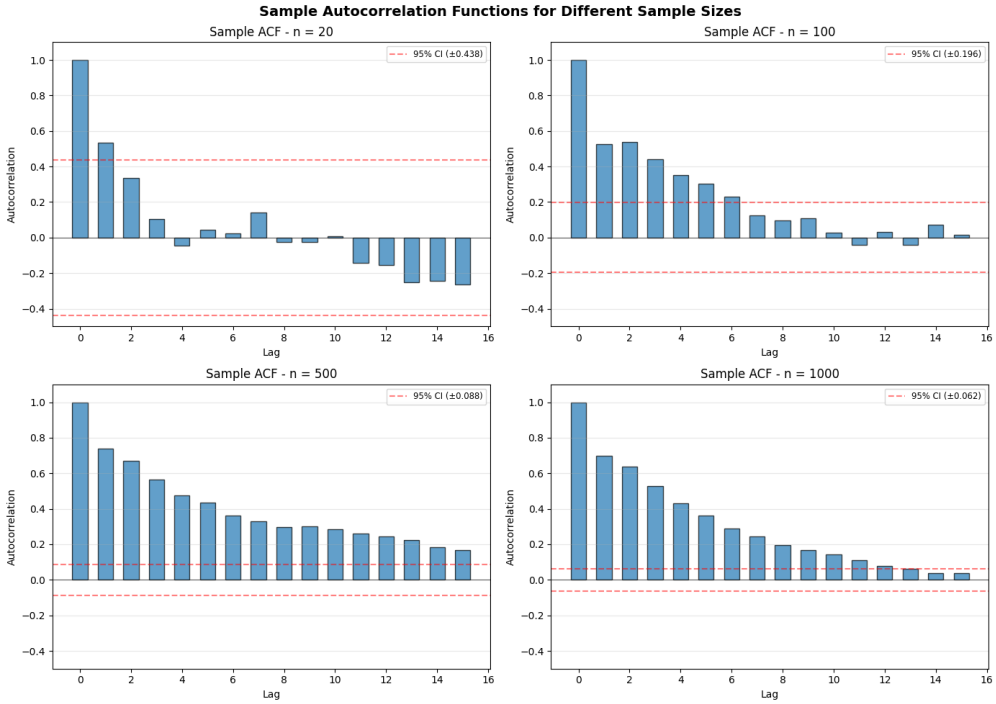

Figure 7: Sample autocorrelation function of the simulated AR(2) process.

Ljung-Box Test Statistics and P-values:

Sample Size	Lag k	Q-statistic	P-value	Reject H0 (α=0.05)
20	1	6.5635	0.0104	Yes
20	2	9.2878	0.0096	Yes
20	5	9.6724	0.0851	No
20	10	10.4176	0.4046	No
100	1	28.4141	0.0000	Yes
100	2	58.7191	0.0000	Yes
100	5	102.2260	0.0000	Yes
100	10	112.1330	0.0000	Yes
500	1	274.3574	0.0000	Yes
500	2	501.7436	0.0000	Yes
500	5	872.0307	0.0000	Yes
500	10	1126.9099	0.0000	Yes
1000	1	487.5471	0.0000	Yes
1000	2	893.8709	0.0000	Yes
1000	5	1491.1105	0.0000	Yes
1000	10	1722.1899	0.0000	Yes

Figure 8: Autocorrelation tests